Gentle microwave treatment of tulip bulbs

Parostok builds on research into microwave seed treatment since 2000. Together with Belgian partners we propose to develop a dedicated machine for chemical-free treatment of living tulip bulbs — designed first of all not to harm the bulb.

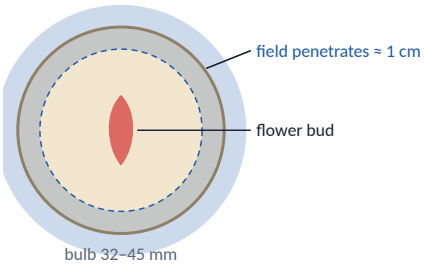
Concept of the line



Functional diagram of the planned machine — not a drawing of an existing unit.

Why a dedicated machine

- A tulip bulb (32–45 mm) is several times larger than the depth a 2.45 GHz field penetrates (about 1 cm).
- In round objects of this size microwave heating can concentrate at the centre — exactly where the flower bud sits.
- Seed equipment therefore cannot simply be reused: bulbs need their own handling, dosing and control.



Target specifications (prototype, draft)

Product	**living tulip bulbs, 10/11 – 12/+ cm**
Prototype throughput	**0.5 t/h (≈ 20 000 bulbs/h)**
Microwave frequency	**2.45 GHz (ISM band)**
Generators	**4–12 × 1.5 kW modules**
Bulb core temperature	**≤ 42 °C, hard limit**
Handling	**single layer, ≈ 1 m wide**
Power supply	**3 × 400 V, 50 Hz**
Target compliance	**CE: (EU) 2023/1230, LVD, EMC**

Values follow the draft technical specification and are fixed only after the bulb tests.

Where we stand

The effect of microwave treatment on tulip bulbs has not been demonstrated yet: we found no published study on living tulip bulbs. The programme therefore starts with bud-safety and pilot forcing trials against an untreated control, and stops if they show no benefit.

Design principles

- Single layer of rotating bulbs, no drops or impacts.
- Dose delivered in sections, with equalisation zones in between.
- **Hard limit: the bulb core never exceeds 42 °C.** Hot-water treatment above about 45 °C is considered harmful to tulip bulbs.
- Optional low-heat pulsed mode.
- Every batch logged: power, temperatures, recipe.
- Easy-to-clean contact surfaces, so that disease is not carried between lots.

What we are looking for in Belgium

- Growers and forcers who can provide bulbs and forcing space for trials.
- An independent research partner for the safety and forcing trials.
- End users and co-funding partners for the prototype.

Staged programme

1	Bulb properties, bud-safety threshold, pilot forcing → go / no-go	6–9 months	**€21–74k**
2	Design and prototype machine (0.5 t/h)	about 2 years	**after gate 1**
3	Independent forcing and field trials	one season	**after phase 2**

Indicative total: up to about €0.5 million. Evidence-based result expected in 2029–2030.

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October 2026. Estimates are indicative and not a commercial offer.